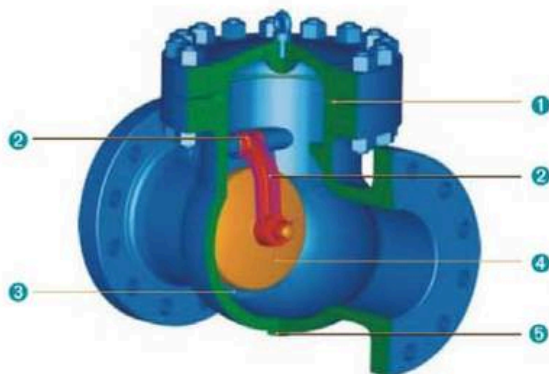


1.3 Swing Check Valve

1.3.1 Swing Check Design Features:



Design description

- BC, Bolted cover
- Swing type, anti-rotation disc
- Seat rings integral with body
- Non-penetrate disc shaft
- Horizontal or vertical service
- Flanged or butt welding ends
- Metallic seating surface

Applicable Standards:

- Steel check valves BS868
- Steel check valves ASME B16.34
- Face to face ASME B16.10
- Ends flanges ASME B16.5
- Butt welding ends ASME B16.25
- Inspection and test API 598

● Seat and Disc for Swing Check :

All parts are accessible from top for easy servicing, welded-in seat is stellite 6 face.

Disc is free to rotate to prevent localized wear
Ground and lapped seat and disc seal surface.

● Body and Bonnet Connection :

The body and bonnet of class 150~900 are usually with studs and nuts, and the body and bonnet of class 1500~2500 check valves are usually of pressure seal design.

● Body to Bonnet Joint :

Stainless steel +flexible graphite wounded gasket is used for class 150~300 check valve; Stainless steel +flexible graphite wounded gasket is used for class 600 check valve and joint gasket is also optional for class 600 check valve; Ring joint gasket is used for class 900 check valve.

2. Flow Coefficient of gate, globe, check valve:

VALVE TECHNICAL DATA-C.V. OF GATE,GLOBE and CHECK VALVE														
DN		50	65	80	100	150	200	250	300	350	400	450	500	600
		2	2-1/2	3	4	6	8	10	12	14	16	18	20	24
GATE	150LB	260	420	625	1150	2650	4850	7750	11500	14000	19000	24000	31000	45000
	300LB	260	420	625	1150	2650	4850	7750	11500	14000	19000	23500	30000	44000
	600LB	260	420	625	1150	2650	4850	7750	11500	13000	18000	22000	27000	40000
GLOBE	150LB	45	72	106	185	440	810	1350	1950	2500	3400	4500	5600	8400
	300LB	45	72	106	185	440	810	1350	1950	2500	3400	4350	5500	8270
	600LB	45	72	106	185	440	780	1300	1880	2390	3200	4060	5040	7520
CHECK	150LB	120	190	285	520	1245	2285	3570	5340	6500	8620	11500	14300	20800
	300LB	120	190	285	520	1245	2285	3570	5340	6500	8620	11500	14300	20800
	600LB	120	190	285	520	1245	2210	3400	5120	6150	8070	10510	12855	18680

3. Inspection and Warranty Of TCX:

Every piece of TCX cast carbon and low alloy steel valves is subjected to 100% pressure tests, according to API 598 requirements. Manufacturer's material test reports and inspection certificates are available on your request, while each valve is guaranteed for 12 months after placement in service, but not exceeding 18 months after shipment from the factories of TCX Corporation. Various tests and inspections of valves made by TCX Corporation include the following. Unless otherwise specified, all TCX cast steel valves shall be subjected to these testing or inspection methods and evaluation criteria.

Test/Inspection Item	Method	Evaluation
Chemical composition analysis		Relevant ASTM Std
Mechanical property test	ASTM A370	Relevant ASTM Std
Pressure tests	API 598	API 598
Radiographic inspection	ASTM E142 / E94	ASME B16.34
magnetic particle inspection	ASTM E709	
Liquid penetrate inspection	ASTM E165	
Low temperature impact test	ASTM E23	Relevant ASTM Std
Dimensional inspection		Relevant Valves Std
Visual inspection		MSS SP-55

Typical TCX Inspection Flow:

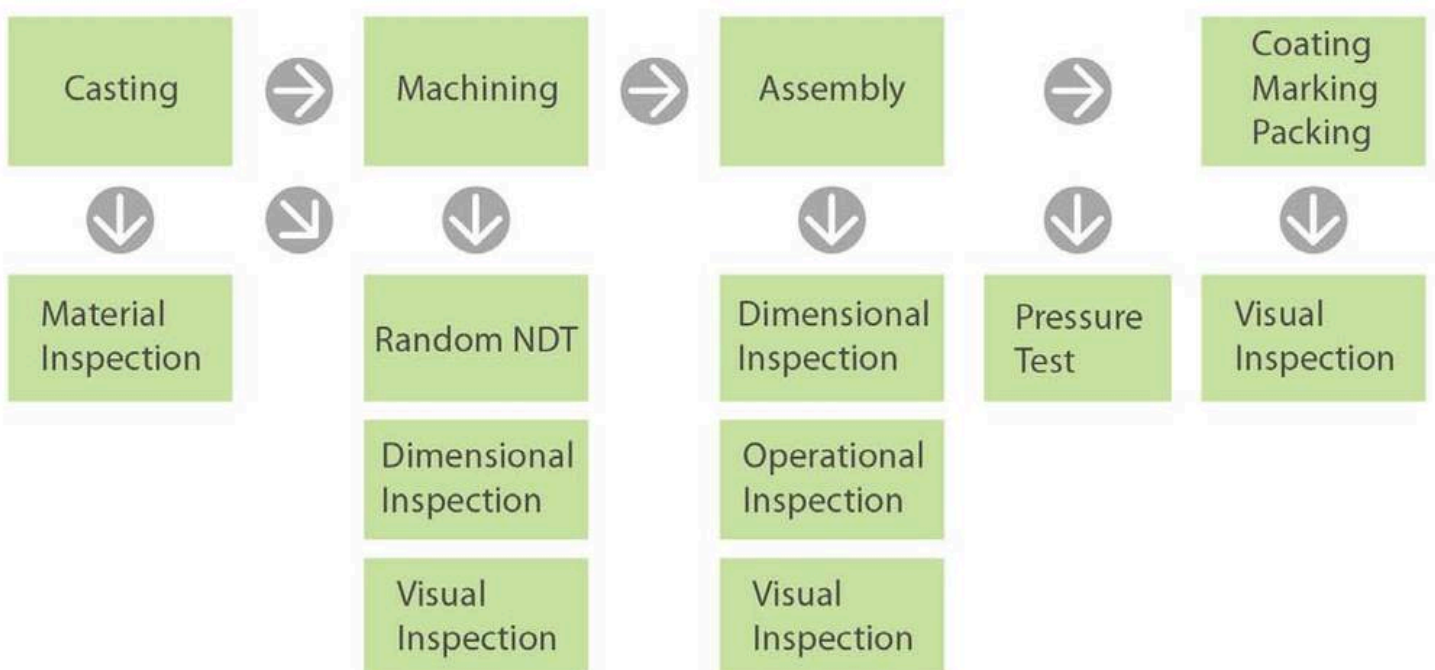


Figure No. CH1RCB8 CH1RCB8-T
CH1BCB8 CH1BCB8-T
CH1JCB8 CH1JCB8-T

FEATURES

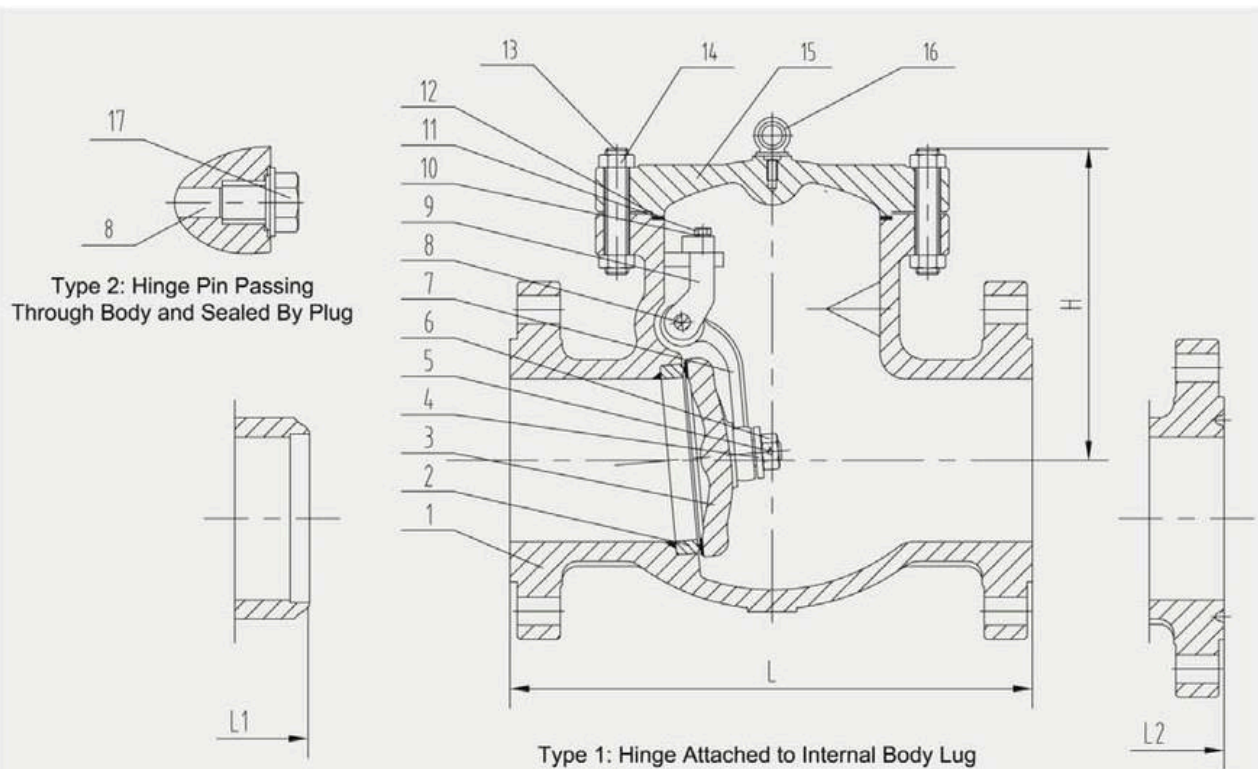
Bolted Cover
Swing Type Disc
Type 1 Standard 2"-10"
Type 2 Standard 12"-30"
6" & Larger With Eye Bolt
By Pass On Request
Lever & Weight Optional

I.D. Marking

TCX SERIES NO.					
ASME B16.34/API 600					
SIZE		BODY			
CLASS		DISC			
100°F		PSI	SEAT		
PSI		*F MAX	PIN		

STANDARD MATERIALS OF PARTS

ITEM	PART NAME	MATERIAL
1	Body	ASTM A216 GR.WCB
2	Seat Ring	ASTM A105 + Setllite Faced
3	Disc	ASTM A216 GR.WCB + 13Cr Faced
4	Disc Washer	ASTM A276 Type 410
5	Disc Nut Pin	Stainless Steel
6	Disc Nut	Stainless Steel
7	Hinge	ASTM A216 GR.WCB
8	Hinge Pin	ASTM A182 GR.F6
9	Bearing Bracket	ASTM A216 GR.WCB
10	Spring Washer	Carbon Steel
11	Hex. Bolt	Carbon Steel
12	Gasket	Soft Iron+Graphite
13	Cover Bolt	ASTM A193 GR.B7
14	Cover Bolt Nut	ASTM A194 GR.2H
15	Cover	ASTM A216 GR.WCB
16	Eye Bolt	Carbon Steel
17	Plug	Carbon Steel



Dimensions and Weights															
SIZE	in	2"	2.5"	3"	4"	6"	8"	10"	12"	14"	16"	18"	20"	24"	30"
	mm	50	65	80	100	150	200	250	300	350	400	450	500	600	750
L/L1	in	8.00	8.50	9.50	11.50	14.00	19.50	24.50	27.50	31.00	34.00	38.50	38.50	51.00	60.00
	mm	203	216	241	292	356	495	622	699	787	864	978	978	1295	1524
L2	in	8.50	9.00	10.00	12.00	14.50	20.00	25.00	28.00	31.50	34.50	39.00	39.00	51.50	-
	mm	216	229	254	305	368	508	635	711	800	876	991	991	1308	-
H	in	6.10	6.70	7.09	8.66	10.55	12.20	14.57	16.73	18.70	20.67	22.83	24.72	34.72	38.27
	mm	155	170	180	220	268	310	370	425	475	525	580	628	882	972
Weight RF	kg	19	26	29	46	77	163	220	366	428	555	775	835	1180	1960

Figure No. CH3RCB8 CH3RCB8-T
CH3BCB8 CH3BCB8-T
CH3JCB8 CH3JCB8-T

FEATURES

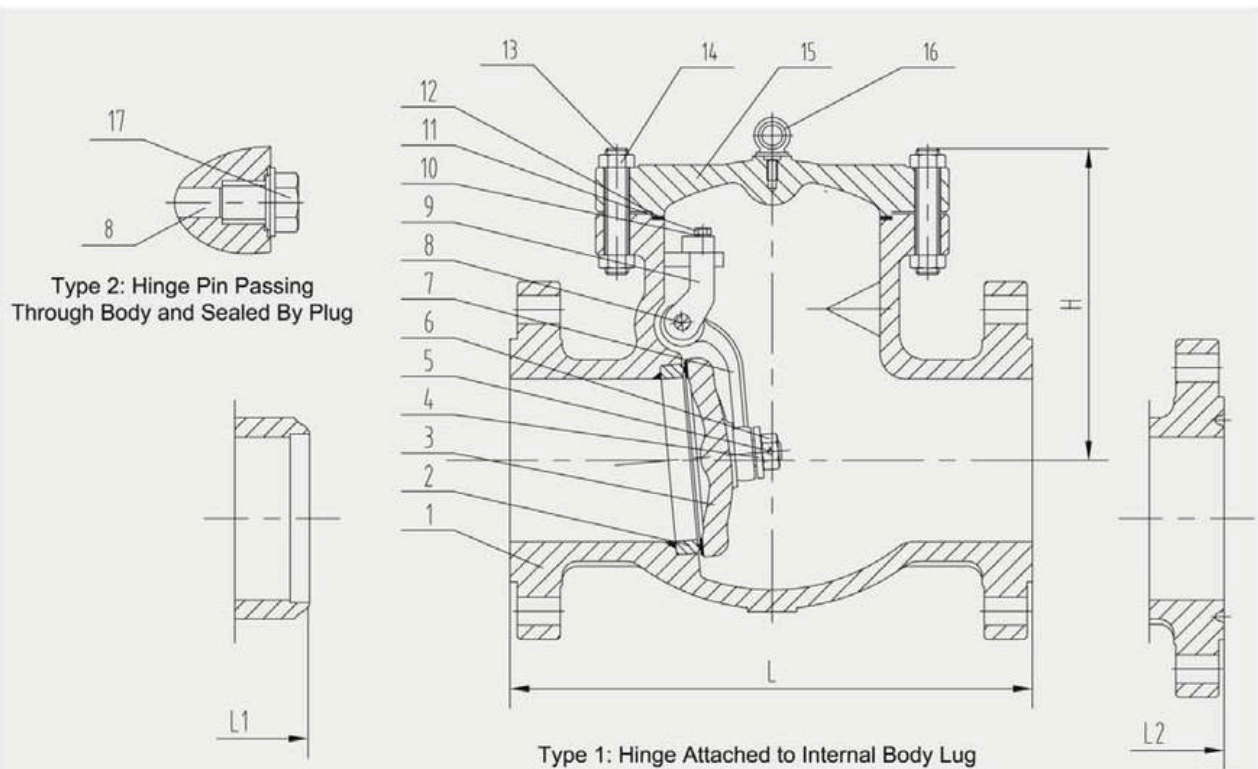
Bolted Cover
Swing Type Disc
Type 1 Standard 2"-8"
Type 2 Standard 10"-24"
3" & Larger With Eye Bolt
By Pass On Request
Lever & Weight Optional

I.D. Marking

TCX		ASME B16.34/API 600	
SERIES NO.			
SIZE		BODY	
CLASS		DISC	
100'/F		PSI	SEAT
PSI		*F MAX	PSI

STANDARD MATERIALS OF PARTS

ITEM	PART NAME	MATERIAL
1	Body	ASTM A216 GR.WCB
2	Seat Ring	ASTM A105 + Setllite Faced
3	Disc	ASTM A216 GR.WCB + 13Cr Faced
4	Disc Washer	ASTM A276 Type 410
5	Disc Nut Pin	Stainless Steel
6	Disc Nut	Stainless Steel
7	Hinge	ASTM A216 GR.WCB
8	Hinge Pin	ASTM A182 GR.F6
9	Bearing Bracket	ASTM A216 GR.WCB
10	Spring Washer	Carbon Steel
11	Hex. Bolt	Carbon Steel
12	Gasket	Soft Iron+Graphite
13	Cover Bolt	ASTM A193 GR.B7
14	Cover Bolt Nut	ASTM A194 GR.2H
15	Cover	ASTM A216 GR.WCB
16	Eye Bolt	Carbon Steel
17	Plug	Carbon Steel



Dimensions and Weights														
SIZE	in	2"	2.5"	3"	4"	6"	8"	10"	12"	14"	16"	18"	20"	24"
	mm	50	65	80	100	150	200	250	300	350	400	450	500	600
L/L1	in	10.50	11.50	12.50	14.00	17.50	21.00	24.50	28.00	33.00	34.00	38.50	40.00	53.00
	mm	267	292	318	356	445	533	622	711	838	864	978	1016	1346
L2	in	11.12	12.12	13.12	14.62	18.12	21.62	25.12	28.62	33.62	34.62	39.12	40.75	53.88
	mm	283	308	333	371	460	549	638	727	854	879	994	1035	1368
H	in	6.89	7.28	7.80	9.25	11.10	13.19	15.15	18.11	20.47	21.81	23.62	26.38	29.53
	mm	175	185	198	235	282	335	385	460	520	554	600	670	750
Weight RF	kg	21	30	39	69	125	210	307	450	680	840	1025	1320	1960

Figure No. CH6RCB8 CH6RCB8-T
CH6BCB8 CH6BCB8-T
CH6JCB8 CH6JCB8-T

FEATURES

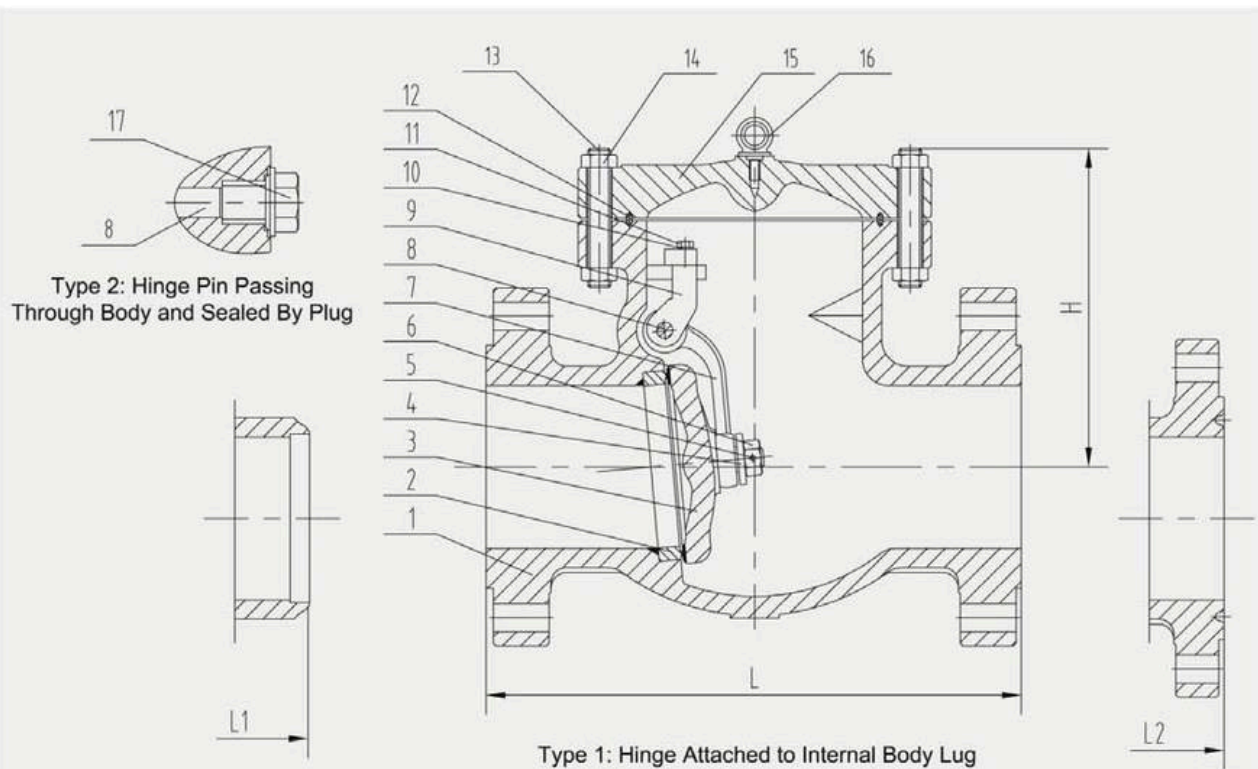
Swing Type Disc
Type 1 Standard 2"-6"
Type 2 Standard 8"-24"
With Eye Bolt
By Pass On Request
Lever & Weight Optional

I.D. Marking

TCX		ASME B16.34/API 600			
SERIES NO.		BODY	DISC	SEAT	
SIZE		PSI	"F MAX	PIN	
CLASS					
100°F					
PSI					

STANDARD MATERIALS OF PARTS

ITEM	PART NAME	MATERIAL
1	Body	ASTM A216 GR.WCB
2	Seat Ring	ASTM A105 + Setllite Faced
3	Disc	ASTM A216 GR.WCB + 13Cr Faced
4	Disc Washer	ASTM A276 Type 410
5	Disc Nut Pin	Stainless Steel
6	Disc Nut	Stainless Steel
7	Hinge	ASTM A216 GR.WCB
8	Hinge Pin	ASTM A182 GR.F6
9	Bearing Bracket	ASTM A216 GR.WCB
10	Spring Washer	Carbon Steel
11	Hex. Bolt	Carbon Steel
12	Gasket, Ring joint	Soft Iron
13	Cover Bolt	ASTM A193 GR.B7
14	Cover Bolt Nut	ASTM A194 GR.2H
15	Cover	ASTM A216 GR.WCB
16	Eye Bolt	Carbon Steel
17	Plug	Carbon Steel



Dimensions and Weights														
SIZE	in	2"	2.5"	3"	4"	6"	8"	10"	12"	14"	16"	18"	20"	24"
	mm	50	65	80	100	150	200	250	300	350	400	450	500	600
L/L1	in	11.50	13.00	14.00	17.00	22.00	26.00	31.00	33.00	35.00	39.00	43.00	47.00	55.00
	mm	292	330	356	432	559	660	787	838	889	991	1092	1194	1397
L2	in	11.62	13.12	14.12	17.12	22.12	26.12	31.12	33.12	35.12	39.12	43.12	47.25	55.38
	mm	295	333	359	435	562	663	790	841	892	994	1095	1200	1407
H	in	7.16	7.87	8.94	10.24	12.80	15.35	19.29	20.79	22.56	25.98	28.35	29.37	37.80
	mm	182	200	227	260	325	390	490	528	572	660	720	746	960
Weight RF	kg	36	49	70	122	269	465	673	875	944	1220	1620	2120	3100